Sounds & Music Cheat Sheet

CS4040 · Unit 4

Assets

Create subfolders called `images`, `sounds` and `music`. Pygame Zero looks in them to find your images, sounds and music.

Folder	Holds	File types
`images`	pictures for Actors	`.png`
`sounds`	sound effects	`.wav` or `.ogg`
`music`	background music	`.ogg` or `.mp3`

Asset names follow the same rules as Python variables: all lowercase, no spaces, only letters, numbers and underscores, and they must begin with a letter. Refer to an asset by its name without the extension: `"alien"`, not `"alien.png"`.

Sounds

Plays shorter `.ogg` or `.wav` audio files. These are for sound effects in your game! Sound functions are called in this format: `sounds.<filename>.<function>()` (docs).

```
sounds.thunder.play()  # plays the sound file called "thunder"
sounds.thunder.play(10)  # plays "thunder" 10 times
sounds.thunder.play(-1)  # plays "thunder" forever on a loop
sounds.thunder.stop()  # stops the "thunder" sound
sounds.thunder.get_length()  # gets the duration in seconds
```

Music

Plays longer `.ogg` or `.mp3` audio files. These are for background music in your game! (docs)

```
music.play("interlude")  # plays file "interlude" on a loop
music.play_once("interlude")  # plays file "interlude" once
music.queue("jam")  # queues "jam" to play after the current track
music.is_playing()  # returns True if music is playing
music.stop()  # stops the music
```

```
music.is_playing("jam")  # returns True if "jam" is playing
music.is_paused()  # returns True if the music is paused
```

```
music.pause()  # pauses the current track
music.unpause()  # resumes the current track
music.fadeout(2.5)  # fades out the current track over 2.5 seconds
music.set_volume(0.5)  # sets volume to 0.5 (0 = silent, 1 = max)
music.get_volume()  # gets the current volume setting
```

Tone

Generates sounds using a built-in synthesizer. `tone.play(pitch, duration)` plays a tone at a certain pitch for a specified length in seconds. `volume` goes from 0 (silent) to 1 (max), default 0.8. `waveform` can be `"sin"` (default), `"square"`, `"saw"` or `"triangle"`.

```
tone.play("E4", 0.5)  # plays an E in octave 4 for 0.5 seconds
tone.play("A#5", 0.25)  # plays an A sharp in octave 5 for 0.25 seconds
tone.play("Bb3", 0.75)  # plays a B flat in octave 3 for 0.75 seconds
tone.play("C4", 1, volume=0.5)  # plays C4 for 1 second at half volume
tone.play("C4", 1, waveform="square")  # plays C4 for 1 second, square wave
```